



ORBITAL JOURNEY
Left, Shubhanshu with crewmates aboard the Crew Dragon spacecraft en route to the ISS, Jun. 25; he smiles despite experiencing space sickness briefly, Jul. 1

Q. How did Earth look like from up there?
What really hit me was its sheer size. You get a very strong sense of your insignificance. If everybody on this planet were brought together—all 8 billion plus people—we'd all probably occupy less than a quarter of Earth's surface. And we are not even the biggest planet in our solar system, not even the biggest solar system in the galaxy, or even the biggest galaxy in the universe. Realising the insignificance of how, who, where and what we are was a humbling experience for me.

Q. What was the colour of Earth from up there?
Mostly blue and white. It is a deep blue, but it keeps changing, depending on the depth of the water, and the angle of the sun. If you are passing a place close to a coast, you see different shades of green and blue. The other thing that struck me were the white clouds, of different shapes and layers. You could see the shadow a towering cloud cast on the lower layers. It was just awesome.

Q. What about the stars?
In terms of distance, we were not too far in space—around 400 km above Earth. But since there is no atmosphere, everything appears with greater clarity. Not bigger, just sharper. But more than the object themselves, what fascinated me was how their geometry changes as you move. We saw the Moon rise and set 16 times every day because we were orbiting Earth every 90 minutes. The best part was when the Moon was about to set, and you see it fade partially. You can still see the Moon because it is above Earth and then gradually it slips beyond Earth. That was one of the most exciting sights for me.

> *"When you are suddenly launched into an environment without gravity, your body rebels. The fluid or blood in your legs, which was earlier flowing down because of gravity, starts going to your head. So, your head starts becoming bigger and your spine elongated"*

Q. When did you get to see India from space?
Only when we got to the International Space Station (ISS). That's because you don't pass over India every time you orbit and, in the Crew Dragon, we crossed India when we were getting some sleep before docking with the ISS. Even on the ISS, I did not get to see India for the first few days as we were tied up with work. Days later, when I was working on an experiment, a NASA astronaut came up to me and told me we were likely to pass over India, and if I'd like to see. I said, of course, and went to the window. It was a night pass, I remember, we were coming from south to north, from the Indian Ocean side. Everything was inky black. Then, suddenly, I saw India, lit up like sparkling diamonds. You could clearly see the outline of the peninsula and make out cities like Mumbai, Delhi and